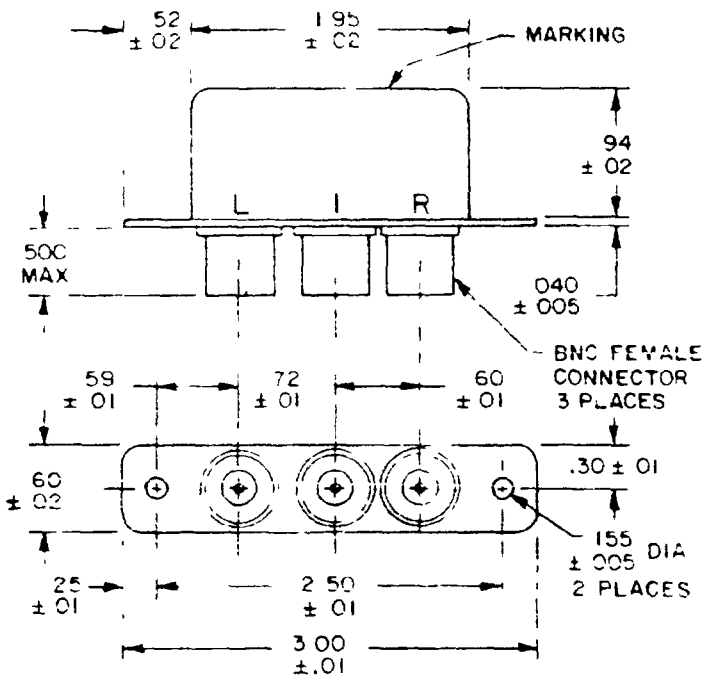
MILITARY SPECIFICATION SHEET

MIXER, DIODE, RADIO FREQUENCY, DOUBLE BALANCED, BNC FEMALE CONNECTORS

This specification is approved for use by all Departments and Agencies of the Department of Defense

The complete requirements for procuring mixers described herein shall consist of this document and the latest issue of Specification MIL-M-28837



INCHES	MM
005	13
01	25
02	51
040	1 02
155	3 94
25	6 35
30	7 62
500	12 70
52	13 21
59	14.99
60	15 24
72	18 29
94	23.88
1 95	49 53
2 50	63 50
3 00	76 20

NOTES
1. Dimensions are in inches
2. Metric equivalents are given for general information only and are based upon 1 00 inch = 25 4 mm

FIGURE 1. Outline drawing for mixers P/N MIL-M28837/6-01N

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REQUIREMENTS

Design and construction
Dimensions and configuration (see figure 1

Electrical characteristics:
Frequency range LO and RF 3.0 - 1000 MHz, IF DC - 1000 MHz
LO input power +7 dBm nominal (+7 dBm used as the test value for the electrical requirements)
Conversion loss (max) 7.5 dB at f_L and f_R 10 MHz-100 MHz, f_I DC - 100 MHz
10 dB at f_L and f_R 3 MHz-1000 MHz f_I DC - 1000 MHz
Noise figure (SSB) (max) 7.5 dB at f_L and f_R 10 MHz-100 MHz, f_I .4 - 100 MHz
10 dB at f_L and f_R 3 MHz-1000 MHz, f_I .4 - 1000 MHz

Isolation in dB (min)

LO-RF	LO-IF	Frequency range
40	40	3 to 100 MHz
30	20	100 to 1000 MHz

Max input power 35 mW (rms)

DC polarity Negative with two in-phase signals applied to the LO and RF ports

Physical and environmental characteristics
Weight 45 grams (1.6 oz) maximum
Temperature
Operating -54° to +100°C
Nonoperating -65° to +100°C.
Shock Test condition C

Part number M28837/6-01X

NOTE. Mixers covered by this specification sheet are substituted for the following manufacturer's part number. This information in no way implies that the manufacturer's part is suitable as a substitute for the military part number

Dash number	FSN	Manufacturer's code	Commercial type
01X	5965-00-566-3068	37695	625467
01Y	5965-00-566-3078	14482	MIA-BNC

Custodians
Army - ER
Navy - EC
Air Force - 85

Review activities
Army - MI
Navy - EC, OS
Air Force - 11, 17, 99
DLA - ES

User activities
Army - AR
Navy - AS, CG, MC
Air Force - 14, 19

Preparing activity
Navy - EC

Agent
DLA - ES

(Project 5965-0986-A1)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
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DOCUMENT IDENTIFIER AND TITLE
MIL-M- 28837/6

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1 HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3 IS THE DOCUMENT RESTRICTIVE?
☐ YES ☐ NO (If "Yes" in what way?)

4 REMARKS

SUBMITTED BY (Printed or typed name and address - Optional)

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DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WH MA BE SEC S/N 0 - 1451R 2

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